

CREDIT CARD FRAUD DETECTION: A HYBRID AUTOENCODER-XGBOOST APPROACH

1. Problem Statement

Credit card fraud detection is a critical challenge for financial institutions due to the highly imbalanced nature of transaction data, where fraudulent transactions account for less than 1% of total records. Traditional machine learning models often fail to detect these rare fraud cases effectively, leading to high false negatives and significant financial loss. The objective of this case study is to design and evaluate a robust fraud detection system capable of accurately identifying fraudulent transactions while handling extreme class imbalance.

2. Objectives

The main objectives of this case study are:

- To reduce false negatives, ensuring fraudulent transactions are not missed
- To improve fraud detection accuracy, with emphasis on recall and AUC
- To handle severe class imbalance using data-level and model-level techniques
- To evaluate the effectiveness of a hybrid anomaly detection and classification model

3. Dataset Description

The experiments were conducted using the Kaggle Credit Card Fraud Dataset, which contains anonymized credit card transactions made by European cardholders.

Dataset Statistics:

- Total transactions: 284,807
- Fraudulent transactions: 492
- Non-fraudulent transactions: 284,315
- Features: PCA-transformed numerical features + Amount + Time
- Target variable: Class (0 = Non-Fraud, 1 = Fraud)

The dataset exhibits extreme class imbalance, making fraud detection a challenging task.

4. Data Preprocessing

To prepare the dataset for effective model training, the following preprocessing steps were performed:

- The Time feature was removed, as it did not provide significant discriminatory information for fraud detection
- The Transaction Amount feature was scaled using standard normalization to handle skewness
- The dataset was split into training and testing sets using stratified sampling to maintain class distribution
- SMOTE (Synthetic Minority Over-sampling Technique) was applied only on the training data to balance the minority (fraud) class

These steps ensured clean, normalized, and balanced data for model training while avoiding data leakage.

5. Model Selection

5.1 Baseline Models

The following baseline machine learning models were implemented for comparison:

- Logistic Regression
- Random Forest
- Standard XGBoost

These models serve as benchmarks to evaluate the effectiveness of the proposed hybrid approach.

5.2 Proposed Hybrid Model

The proposed model integrates unsupervised anomaly detection with supervised classification:

- An Autoencoder is trained exclusively on non-fraudulent transactions to learn normal transaction patterns
- The model reconstructs input transactions and computes reconstruction error, which serves as an anomaly score

- The anomaly score is fused with selected transaction features
- The enriched feature set is passed to an XGBoost classifier for final fraud prediction

This hybrid approach leverages both anomaly detection and ensemble learning strengths.

6. System Architecture

The system consists of four major components:

- Data preprocessing and normalization layer
- Hybrid feature selection layer
- Autoencoder-based anomaly detection module
- XGBoost-based fraud classification module

The anomaly score generated by the autoencoder is combined with selected features before classification.

7. Results and Performance Comparison

Performance Metrics Summary:

Model	Precision	Recall	F1-Score	AUC
Logistic Regression	Low	Moderate	Moderate	0.89
Random Forest	Better	Better	Better	0.94
XGBoost	High	High	High	0.97
Hybrid AE + XGBoost	Very High	Very High	Very High	0.99

The hybrid model significantly outperforms baseline models, particularly in recall and AUC, which are critical for fraud detection.

8. Key Insights

- Autoencoder-based anomaly detection improves recall by identifying rare fraudulent patterns
- Feature fusion enhances model robustness and generalization
- The hybrid approach reduces false positives, improving operational efficiency
- Precision-Recall analysis confirms strong minority class learning

- The proposed model achieves 99% AUC, demonstrating excellent discriminative ability

9. Recommendations

Based on the case study findings:

- Financial institutions should deploy hybrid fraud detection models that combine anomaly detection with supervised learning
- Continuous monitoring is essential to detect model drift and evolving fraud patterns
- Periodic retraining with recent transaction data is recommended
- Explainable AI techniques should be integrated to improve trust and regulatory compliance
- Real-time inference capabilities should be implemented for immediate fraud alerts

10. Conclusion

This case study demonstrates that combining autoencoder-based anomaly detection with XGBoost classification significantly enhances credit card fraud detection performance. The proposed hybrid model effectively addresses class imbalance, improves recall, reduces false positives, and is well-suited for real-world financial applications. With an AUC of 0.99 and superior recall metrics, the system proves to be a reliable solution for detecting fraudulent transactions while minimizing false alarms in financial institutions.